

Caffeine enhances endothelial repair by an AMPK-dependent mechanism.



OBJECTIVE: Migratory capacity of endothelial progenitor cells (EPCs) and mature endothelial cells (ECs) is a key prerequisite for endothelial repair after denuding injury or endothelial damage. **METHODS AND RESULTS:** We demonstrate that caffeine in physiologically relevant concentrations (50 to 100 micromol/L) induces migration of human EPCs as well as mature ECs. In patients with coronary artery disease (CAD), caffeinated coffee increased caffeine serum concentration from 2 micromol/L to 23 micromol/L, coinciding with a significant increase in migratory activity of patient-derived EPCs. Decaffeinated coffee neither affected caffeine serum levels nor migratory capacity of EPCs. Treatment with caffeine for 7 to 10 days in a mouse-model improved endothelial repair after denudation of the carotid artery. The enhancement of reendothelialization by caffeine was significantly reduced in AMPK knockout mice compared to wild-type animals. Transplantation of wild-type and AMPK(-/-) bone marrow into wild-type mice revealed no difference in caffeine challenged reendothelialization. ECs which were depleted of mitochondrial DNA did not migrate when challenged with caffeine, suggesting a potential role for mitochondria in caffeine-dependent migration. **CONCLUSIONS:** These results provide evidence that caffeine enhances endothelial cell migration and reendothelialization in part through an AMPK-dependent mechanism, suggesting a beneficial role for caffeine in endothelial repair.